SIDInspector: A Mapping-First Diagnostic Resource for Semantic-ID Tokenizers

Jiandong Ding, Heng Chang, Huijie Qin, and Tianying Liu
{dingjiandong2,heng.chang,qinhuijie,liutianying2}@huawei.com
Huawei Technologies
Shanghai, China

Abstract

Semantic-ID (SID) tokenizers are increasingly reused as standalone artifacts in generative recommendation: an exported item-to-code mapping becomes the address space that a later sequence generator must use. These mappings rarely come with a common inspection interface, so coverage gaps, full-code aliasing, behaviorally weak prefixes, tail compression, and prefix fan-out are often found only after downstream training. We present SIDInspector, a mapping-first diagnostic resource for SID tokenizer artifacts. SIDInspector defines a small adapter contract over item mappings, metadata, interactions, and optional generator traces; validates the contract; and reports mapping-level probes for utilization, aliasing, neighborhood alignment, popularity allocation, and structural cost, with hooks for temporal churn and generator traces.

SIDInspector reports inspectable artifact profiles before downstream leaderboard scores. The released resource covers four tokenizer artifact lines: a same-item GRID/RQ-KMeans-style and ReSID/GAOQ contrast on 23,742 Musical items, plus released LETTER and LC-Rec item-index artifacts. In the Musical contrast, the GRID-style feature-text export has 3,749 unique full codes and a 0.977 full-code aliasing rate, while ReSID/GAOQ is aliasing-free in its exported mapping. Yet the strongest prefix–co-occurrence alignment comes from a deterministic category-prefix control, not from either learned export row (0.447 versus 0.154 and 0.055–0.080), showing that addressability and behaviorally meaningful prefixes should be inspected separately. Cross-domain, fixed-reranker, and mechanism-probe checks support the same diagnostic direction: prefix alignment is a candidate-exposure signal, while final ranking quality remains a downstream model question.

CCS Concepts

• **Information systems → Recommender systems**; *Retrieval models and ranking*.

Keywords

semantic IDs, generative recommendation, tokenizer artifacts, diagnostic evaluation, recommender systems

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1 Introduction

Semantic-ID (SID) tokenization has become a practical artifact boundary for generative recommendation. Systems such as TIGER map items into discrete code sequences so that a sequence model can generate candidate identifiers rather than score every catalog item directly [22]. Once exported, that item-to-code mapping becomes the address space exposed to the generator. Recent work has expanded this space quickly: residual quantization frameworks, recommender-native encoding, collaborative tokenization, differentiable tokenization, non-uniform capacity, qualified collisions, variable-length interfaces, collision-aware evaluation, and drift-aware refreshes all change the artifact that downstream users must inspect before generator training [4, 7, 8, 11, 14, 18, 27, 32, 34]. This paper treats those exports as a shared artifact interface: the goal is not to reimplement the literature, but to make item-to-code mappings inspectable under the same contract.

Reusable tooling has not caught up with this artifact boundary. A new tokenizer repository may expose checkpoints, item mappings, codebooks, or only intermediate feature files. Downstream users then need to answer basic engineering questions before training a generator: do all catalog items have valid codes, are many items aliased to the same full code, do prefixes reflect user co-occurrence rather than only metadata taxonomy, are tail items allocated enough address capacity, and does the prefix structure create large fan-out? Today these checks are usually reimplemented ad hoc inside each paper. The resulting evidence is difficult to compare because each method reports a different mixture of codebook usage, downstream ranking, and qualitative examples.

Aggregate ranking metrics remain necessary, but they are the wrong interface for inspecting the exported address space itself. Prior recommender tooling has argued for behavioral tests beyond NDCG-style aggregates [5]; SID tokenizers add a more concrete resource problem: the item mapping should be validated and profiled before a full generator consumes GPU time. Underused codebooks, repeated full codes, behaviorally weak prefixes, tail compression, and large prefix fan-out can be hidden by a downstream score, yet they are direct properties of the reusable tokenizer artifact.

We present SIDInspector, a diagnostic/interface resource for inspecting item-to-SID tokenizer artifacts before or alongside downstream recommendation evaluation. SIDInspector is deliberately neither a new tokenizer nor a RecBole-style coverage benchmark. It asks a narrower and reusable question: given any tokenizer artifact that can emit item-level code sequences, what can be said about its capacity use, aliasing, behavioral alignment, head-tail allocation, and structural cost before retraining a full generator?

The resource contribution is threefold. First, SIDInspector defines a small adapter contract for SID assignments, item metadata,

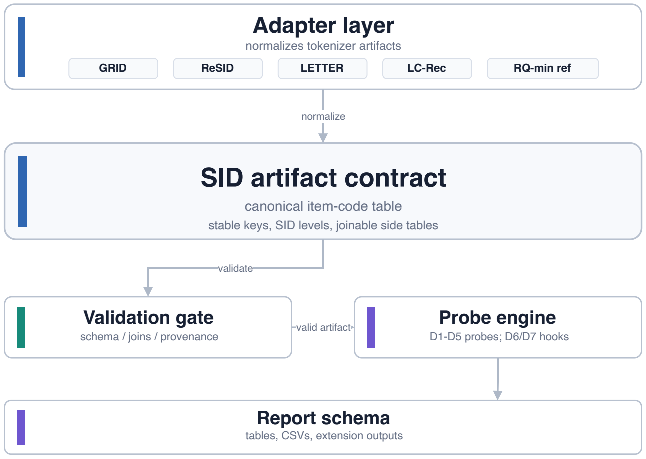

Figure 1: SIDInspector architecture.

interaction histories, and optional generator outputs, plus validators that reject incomplete or ambiguous artifacts before metrics are reported. Second, it implements D1–D5 mapping-level probes for utilization, aliasing, neighborhood alignment, popularity allocation, and structural cost, with D6 churn support for continual-tokenizer settings. Third, it provides worked examples across four tokenizer artifact lines: GRID/RQ-KMeans-style and ReSID/GAOQ form the same-item diagnostic contrast, while LETTER and LC-Rec test released item-index adapters. References, controls, and mechanism probes calibrate the diagnostics. The central finding is narrow but useful: learned addressability and behavioral prefix alignment diverge, so they should be inspected as separate artifact properties before method leaderboard evaluation [6].

2 Artifact Interface and Validation

SIDInspector treats a tokenizer export as an inspectable resource artifact, not as an opaque component of a trained recommender. The minimum input is a table `sid_assignments(item_id, sid_0, ..., sid_L)`. Optional tables provide metadata, interactions, refresh pairs, and generator outputs. This mapping-first contract lets adapters normalize heterogeneous repositories while keeping the inspection independent of a particular training loop. Figure 1 summarizes the architecture and artifact boundary.

Artifact contract. The required contract has three checks: each row has a stable item key, every SID level is a discrete token, and the diagnostic item universe is joinable to metadata and interaction slices. Metadata enables semantic/category slices; interactions enable collaborative-neighborhood and head-tail diagnostics. The `generator_outputs` table is optional because many public tokenizer releases expose item codes but no trained generator traces.

Adapter specification. An adapter is a thin export layer, not a retraining recipe. It must emit `sid_assignments` with stable item keys, a method/dataset label, and one column per SID level. If available, it also emits metadata, interactions, refresh-pair snapshots, or generator traces using the same item keys. The validator checks key uniqueness, missing mappings, depth consistency, join coverage, and source provenance before any row can enter the diagnostic tables. The minimal schema is deliberately small:

$$\texttt{item\_id}, \texttt{method}, \texttt{dataset}, \texttt{sid\_0}, \ldots, \texttt{sid\_L} \rightarrow \{\text{D1}, \ldots, \text{D5}\}.$$

Interactions add D3/D4 slices; paired mappings add D6; generator traces add D7. This makes a new tokenizer easy to attach without rewriting D1–D5. Current D1–D5 reports use a rectangular code table; variable-length tokenizers can be attached by exporting padded or masked levels plus realized length, but variable-length serving behavior remains structural D5 evidence unless generator traces are available.

Terminology. A *diagnostic probe* is a named metric family: D1 utilization, D2 aliasing, D3 neighborhood alignment, D4 popularity allocation, D5 structural cost, D6 temporal churn, or D7 generation traces. A *controlled mechanism probe* is an input row designed to activate one known mechanism and check whether the diagnostic probe reacts. The main paper tables use D1–D5. D6 is implemented and released as an extension example for paired refresh mappings; D7 is only an interface hook until generator outputs or beam traces are supplied.

Claim scope. SIDInspector reports tokenizer artifact rows, reference rows, and mechanism probes under the same schema. The current tokenizer lines are GRID-style RQ-KMeans, ReSID/GAOQ, LETTER, and LC-Rec. The first two support the same-item diagnostic case study; the latter two test released item-index adapters. References and probes calibrate the diagnostics rather than increasing tokenizer coverage. Optional rows exercise extensions such as drift or non-Amazon schemas, while literature-only methods motivate future adapters.

Validation before metrics. Adapter outputs are interpreted only after checks for item-count agreement, missing codes, duplicate keys, depth consistency, and provenance. This guard is necessary because public SID repositories may expose checkpoints, item mappings, codebooks, intermediate feature files, or only partial releases. Once the normalized tables pass validation, D1–D5 can be compared across artifacts.

Prior recommender resources such as RecList and Elliot motivate diagnostic and reproducible evaluation beyond a single aggregate metric [1, 5]. SIDInspector applies that resource logic to a newer artifact boundary: the item address space consumed by generative recommenders.

3 Diagnostic Probes and Artifact Catalog

SID papers now optimize different artifact properties. Learned item indexing and TIGER-style generative retrieval established the interface [12, 22]. Later systems diversify how codes are learned and used: collaborative or learnable tokenization [19, 25, 33], behavior-semantic or multimodal evidence [3, 26], collision and head-tail objectives [11, 20, 30], and variable-length, drift, or deployment-facing SID designs [2, 4, 15] all expose different inspection needs. The literature space is broader than the artifacts that SIDInspector currently executes. Table 1 therefore catalogs the released resource surfaces rather than repeating the experimental rows. The quantitative diagnostic evidence appears in Tables 2 and 3; Table 1 tells a reviewer what can be inspected in the artifact and where to start.

Table 1: Released resource surfaces.

Resource part	Reviewer-facing check	Entry point
Adapter contract	New tokenizer exports can be normalized into stable item-code rows with optional side tables.	README; template
Validation gate	Keys, SID levels, joins, and provenance are checked before diagnostics run.	package verifier
D1–D5 probes	Valid artifacts produce utilization, aliasing, alignment, allocation, and cost reports.	package verifier; CLIs
Released adapters	LETTER and LC-Rec item-index files enter the same contract as local exports.	adapter registry
Evidence index	Paper-table numbers are linked to compact snapshots and commands.	reproducibility matrix
Calibration assets	Control rows and mechanism probes check that D2, D4, and D5 respond to known stressors.	evidence snapshots
Extension hooks	D6 churn, downstream probes, and D7 trace inputs expose the extension path.	README; CLIs

Notation. Let $\mathcal{I}$ be the inspected item set and let a tokenizer export a code sequence $z(i) = (z_1(i), \ldots, z_L(i))$ for each item i. We write the level-ℓ prefix as $p_\ell(i) = (z_1(i), \ldots, z_\ell(i))$ and the full alias set for code c as $A(c) = \{i \in \mathcal{I} : z(i) = c\}$. The probes below operate on z, with metadata and interactions added only when a diagnostic needs semantic or collaborative slices.

D1–D5 mapping-level probes. D1, *utilization*, reports per-level usage, prefix counts, imbalance summaries, and dead or underused code indicators. D2, *aliasing*, measures the fraction of items whose full SID belongs to a non-singleton code bucket, plus the same quantity for prefixes. This is an item-in-alias-set profile rather than the number of duplicate code values; causal harm requires intervention or downstream exposure checks beyond this mapping profile. A bounded interaction-qualified mechanism probe is included because collision-aware work argues that duplicate-code assignments are not uniformly harmful, and recent tokenizer-comparison work shows that SID-level evaluation can diverge from item-level recommendation under full-code collisions [11, 32]. D3, *neighborhood alignment*, asks whether prefix neighborhoods recover train-only item co-occurrence neighbors. For each user, SIDInspector forms item pairs among bounded train interactions, ranks neighbors by co-occurrence count, keeps top-k directed neighbors per item, and reports the fraction whose neighbor shares $p_\ell(i)$; the weighted score averages edges, while the mean score averages items. Metadata purity is auxiliary context, not collaborative quality. D4, *popularity allocation*, splits items by popularity and measures whether full-code capacity and prefix structure are allocated differently across head, mid, and tail items. D5, *structural cost*, reports SID length, prefix fan-out, duplicate codes, and trie-like expansion pressure. Measured serving latency depends on the generator and decoding stack; long, variable-length, or asymmetric SID methods make that distinction necessary [4, 9, 13, 24, 28, 31].

Extension probes. D6, *temporal churn*, computes mapping changes between tokenizer refreshes for drift-aware settings [2, 7]. It is implemented as an optional refresh-pair extension. D7, *generation traces*, covers invalid paths, next-token entropy, and duplicate generated candidates. D7 requires generator-output tables or beam traces; the current evidence focuses on the mapping layer.

Table 2: Musical diagnostic profile.

Artifact	Unique SIDs	D2	D3	D4	D5 levels
RQ-style exports					
GRID-style ft	3,749	0.977	0.055	0.370	64/3.4k/3.7k
GRID-style cap	9,874	0.779	0.080	0.639	32/9.3k/9.9k
RQ-min ref	17,247	0.440	0.065	0.883	32/2.4k/17.2k
ReSID/GAOQ export					
ReSID†	23,742	0.000	0.154	1.000	32/1.3k/23.7k
Controls					
Cat-prefix†	23,742	0.000	0.447	1.000	30/83/313/23.7k
Pop-balanced	22,707	0.086	0.303	0.962	4/1.0k/22.7k/22.7k
Hash-collide	256	1.000	0.004	0.032	256/256/256/256

Table 3: Mechanism-probe calibration.

Mechanism	D	Baseline signal	Activated signal
Qualified aliasing	D2	hash 1.19x	co-occur 3.86x
Capacity budget	D1/D4	head 1.000	tail 0.028
Variable depth	D5	max-depth 12,010	active 7,914

4 Worked Examples and Probe Evidence

SIDInspector is evaluated through interface coverage and diagnostic separation. The evidence has two layers. First, a same-item Musical contrast compares GRID-style RQ-KMeans and ReSID/GAOQ under one adapter contract, which is the right setting for the addressability-versus-alignment finding. Second, LETTER and LC-Rec exercise released item-index artifacts, showing that the same contract is not tied to the Musical case study. Controls and mechanism probes then separate structural, behavioral, and capacity-driven signals.

Worked-example protocol. The case study uses the same 23,742 Musical items for the GRID and ReSID rows. GRID is a controlled feature-text export: ReSID processed feature text is embedded and passed through a released GRID/RQ-KMeans-style residual k-means module. ReSID is a bounded FAMAE-to-GAOQ export on the same item universe. The protocol supports a same-item diagnostic contrast; end-to-end pipeline reproduction is outside this artifact-layer protocol. In Table 2, D2 is full-code aliasing, D3 is level-1 weighted co-occurrence-prefix recall, D4 is tail unique-SID ratio, and D5 lists active prefix counts by depth. Daggered rows are structurally item-unique; bold rows mark the primary same-item contrast and italics mark controls.

Table 2 makes the same-item rows comparable under one schema. The GRID and ReSID rows are the tokenizer contrast; the capacity ablation, RQ-min reference, and controls explain which parts of the contrast come from capacity, addressability, or prefix semantics. In this setup, GRID-style ft exposes high aliasing pressure and limited tail capacity, while the ReSID/GAOQ row exports one full code per item and has higher D3-L1 collaborative prefix recovery. The daggered rows mark a structural floor: collision-free full codes capture addressability, while recommendation quality depends on downstream generation and ranking.

The capacity-matched GRID row tests whether the original GRID pressure is only a budget artifact. With a larger 32/1280/1280 budget, unique full codes rise to 9,874 and D2 aliasing drops to 0.779, but the row remains far from the structurally item-unique ReSID and category-prefix rows. Capacity explains part of the pressure, while the non-item-unique leaf keeps the contrast at the artifact-profile layer. RQ-min is a minimal residual-quantization reference adapter that passes the same validator; it tests adapter independence on the full Musical item universe and is reported as a reference row rather than an additional tokenizer line.

Diagnostic findings. The worked example supports one central finding: addressability is not behavioral prefix alignment. ReSID is structurally item-unique, and the category-prefix control is also alias-free, but D3 separates their profiles. ReSID is a learned export, whereas the category row is a deterministic control for interpreting the metric scale. The non-learned category-prefix row recovers more level-1 co-occurrence neighbors (0.447) than ReSID (0.154) or either GRID row (0.055–0.080). The result locates a prefix-alignment gap: learned or item-unique addresses can still fail to group behaviorally related items. This failure mode is visible only when the mapping is inspected directly.

The supporting evidence is diagnostic separation. Across the RQ-style rows, D2 aliasing falls from 0.977 (GRID-style ft) to 0.779 (GRID-style cap) and 0.440 (RQ-min), while D3 stays in the narrow 0.055–0.080 range. Capacity and aliasing can improve without producing behaviorally aligned prefixes, making D2 and D3 complementary diagnostics. The ablation weakens a simple "too little capacity" objection without making the row item-unique. Table 3 gives the same point as controlled checks: qualified aliasing separates co-occurrence aliases from hash aliases, capacity probes separate head and tail allocation, and variable-depth probes separate maximum-depth from realized prefix cost.

Released item-index adapters test the resource beyond the same-item case. LETTER and LC-Rec Instruments artifacts both pass the same validation path on 9,922 items; both expose 9,897 unique full SIDs, with D3-L1 0.109 for LETTER and 0.052 for LC-Rec. They appear in the catalog rather than the same-item table because they use a different item universe; their role is to show that SIDInspector ingests released tokenizer mappings as well as locally generated contrasts.

Additional checks bound the D3 claim and are indexed in the released reproducibility matrix. On an All_Beauty 20k vertical, a coarse category-prefix control reaches D3-L1 0.968 while GRID/RQ-KMeans feature-text rows are 0.081, 0.087, and 0.090 across seeds 42–44. Here D3 measures taxonomy–co-occurrence alignment under the available coarse category signal, which differs across datasets. The high All_Beauty value therefore remains a diagnostic signal: the available coarse taxonomy is more aligned with observed co-occurrence there than in Musical. Two unified 5,000-user fixed-reranker probes sharpen the boundary. Across eight Musical rows and six All_Beauty rows, D3 is strongly associated with candidate target recovery under train-only prefix retrieval (depth-1 Spearman 0.976 and 1.000). At $K = 20$, the corresponding Spearman values with ranked Recall/NDCG are 0.970/0.952 on Musical and 0.657/0.657 on All_Beauty. D3 exposes candidate-exposure structure before the reranker or generator decides final ordering. A Sports 20k GRID export has complete metadata and interaction joins, D3-L1 0.055, duplicate-SID rate 0.592, and D5 prefixes 128/7986/8165. Sports, MovieLens, DACT, LETTER, and LC-Rec rows exercise portability and adapter coverage around the same-item Musical finding.

5 Artifact Availability and Limits

The released repository contains the SIDInspector Python package, adapter template, metric runners, sample inputs, tests, documentation, an MIT license, and a clean-checkout verifier. The released resource is available at https://github.com/jdding/sidinspector under tag `sidinspector-cikm2026-resource-v0.6`. It is intended as a reusable diagnostic resource: a user can clone the package, run the toy diagnostic, validate a new item-to-SID mapping, and obtain D1–D5 reports without experiment caches. The released repository focuses on the reusable interface and diagnostic implementation; selected experimental evidence is reported in the paper.

The resource supports three modes. A downstream recommender researcher can use the diagnostics as a pre-training triage step: artifacts with missing items, inconsistent depth, severe aliasing, or weak collaborative-prefix recovery are flagged for inspection before generator training consumes GPU time. A method author can add an adapter that emits the normalized SID-assignment table, then receive D1–D5 reports without changing their training loop. A reader can run the clean-check verifier against the reusable package. These signals complement Recall@K and NDCG by making tokenizer artifacts inspectable before a full benchmark or generator-training run.

The evidence supports a resource-interface claim over four tokenizer lines: GRID-style RQ-KMeans, ReSID/GAOQ, LETTER, and LC-Rec. RQ-min and the controlled stressors are reference rows for interpreting the diagnostics; CARD enters the catalog once generated codes can be joined and validated. D2 reports mapping aliasing; D3 uses prefix-neighborhood and reranker checks; D5 measures structural cost; D6 is available for paired refresh mappings; D7 activates when generator traces are supplied.

New adapters emit the same normalized tables and record feature or checkpoint provenance. A tokenizer line enters the catalog when item-level codes pass the validator and join to metadata or interactions; mechanism probes remain diagnostic calibration rows. Future versions should add causal collision-harm validation, broader adapter coverage, generator-output diagnostics, and cross-domain or unified search/recommendation checks [10, 17, 21]. Industrial SID deployment and ranking studies further motivate treating these artifact properties as engineering objects [15, 16, 23, 29].

GenAI Usage Disclosure

Generative AI tools assisted with language editing and code debugging. All experimental results were produced by author-written scripts and verified against saved artifacts.

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